Real-World Accuracy Assessment of

Latency-Resilient Cooperative MARL for Adaptive Traffic Signal Control

A Critical Evaluation of the Released Thesis

Against Real-World Deployment Conditions

Independent Technical Review

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Abstract

This report provides a rigorous, evidence-based assessment of how accurately the released thesis on latency-resilient cooperative Multi-Agent Reinforcement Learning (MARL) for adaptive traffic signal control reflects real-world conditions. Unlike the comparative version analysis (`overall.pdf`), which tracked document evolution, this assessment evaluates the *final released document* (`version3.pdf`) against established empirical data from vehicular communication networks, traffic engineering practice, edge computing platforms, and deployed intelligent transportation systems (ITS).

Each claim is classified as ✓ **adequately accurate**, **!** **moderately concerning**, or **X** **critically inaccurate** with respect to real-world conditions. The assessment reveals that while the theoretical framework is sound, several deployment-critical claims lack empirical grounding, and the thesis's characterization of its own contribution scope requires significant clarification.

Contents

1 Executive Summary

This section presents a consolidated accuracy scorecard across all assessed domains, followed by the single most critical finding that pervades the entire document.

Table 1. Overall Real-World Accuracy Scorecard

Assessment Domain	Rating	Score	Primary Issue
Communication Parameter Ranges	!	5/10	C4 unrealistic; C2–C3 partially aligned
V2X Technology Characterization	!	4/10	Superficial treatment; no protocol-stack detail
Traffic Simulation Fidelity	X	3/10	TRAFI/SAFE confusion; no real data calibration
Network Topology Realism	!	4/10	Idealized grid; no real road geometry
Hardware Architecture Claims	X	2/10	Purely conceptual; no specifications
Safety & Reliability Guarantees	!	4/10	Theoretical assertions only; no verification
Mathematical Formalization	✓	7/10	Sound DEC-POMDP extension with AoI
Empirical Validation Status	X	1/10	No actual experiments conducted
Reward Design Practicality	!	5/10	Unvalidated cooperative weighting
Scalability Claims	!	3/10	3×3 grid only; no scaling evidence
Weighted Aggregate	!	**3.8 / 10**	

! Key Finding — Presentation of Projections as Results

The most significant finding of this assessment is that the thesis presents a **simulation design intended for future validation** (Chapter 4, § 4.1: "*intended for future empirical validation*") yet discusses outcomes in Chapter 5 as if experiments had been conducted. This creates a fundamental accuracy gap: the "results" are *theoretical projections*, not empirical findings. Any reader interpreting Chapter 5 as validated experimental data would be systematically misled.

2 Assessment Methodology

2.1 Evaluation Framework

Each claim in the released thesis is evaluated against three categories of real-world reference material:

(a) **Empirical measurements** from published vehicular network field trials (e.g., USDOT V2X Pilots in Tampa and Wyoming; European C-ITS Corridor deployments).

(b) **Engineering standards** including IEEE 802.11p, 3GPP Release 16/17 for C-V2X, NTCIP 1210 for traffic signal controllers, and the SUMO simulation reference documentation.

(c) **Deployed system reports** from SCATS, SCOOT, and emerging AI-based systems (e.g., Surtrac in Pittsburgh; SMART Signal in Minnesota).

2.2 Accuracy Rating Definitions

Table 2. Rating Criteria and Definitions

Rating	Definition
✓ **Adequate**	Claim aligns with published empirical data within reasonable bounds (± 20 %) or correctly represents the state of the art with appropriate caveats.
! **Moderate**	Claim is partially correct but oversimplifies, omits critical qualifiers, or uses values outside typical empirical ranges without justification.
X **Critical**	Claim contradicts established empirical data, misrepresents a technology's capability, or presents theoretical results as empirical findings.

3 Communication Model Accuracy

3.1 Latency Parameter Ranges

The thesis defines four communication configurations in Table 4.2. Table 3 compares each against published field data.

Table 3. Thesis vs. Real-World Latency Ranges

Config.	Thesis τ (ms)	Real-World (ms)	Reference Context	Rating
C1 (Ideal)	$[0,\ 0]$	$[5,\ 20]$	Local fiber LAN	!
C2 (Moderate)	$[20,\ 100]$	$[30,\ 150]$	DSRC urban	✓
C3 (Severe)	$[100,\ 500]$	$[100,\ 300]$	DSRC interference	!
C4 (Extreme)	$[500,\ 2000]$	N/A	Infrastr. failure	X

? C1 — Zero-Latency Idealization Is Unrealistic

Even with fiber-optic backhaul between traffic controllers, non-zero latency arises from processing overhead, protocol serialization, and switch propagation. Field measurements from the USDOT Connected Vehicle Pilot in Tampa report minimum V2I latency of **8–15 ms** even under ideal conditions. The thesis should adopt a small but non-zero baseline, e.g. $\tau \in [5, 20]$ ms.

! C4 — Does Not Represent a Credible Operational Scenario

A latency of 500–2000 ms with 50 % packet loss corresponds to **near-total communication infrastructure failure**—a scenario in which no cooperative system (MARL or otherwise) would be expected to function. In real-world deployments such conditions trigger *immediate fallback* to local actuated control, not graceful degradation of a cooperative algorithm.

Including C4 inflates the apparent benefit of the resilience mechanisms by benchmarking against an unrealistic worst case. A more honest extreme scenario would be $\tau \in [200, 500]$ ms, $p_{\text{loss}} = 20\,\%$, representing severe interference from construction, weather, or RF congestion.

3.2 Packet Loss Rate Accuracy

Table 4. Thesis vs. Real-World Packet Loss Rates

Config.	Thesis p_{loss}	Empirical Range	Context	Rating
C1	0.00	0.001–0.01	Fiber / fixed link	!
C2	0.10	0.05–0.15	DSRC urban, 300 m	✓
C3	0.30	0.15–0.25	DSRC dense urban	!
C4	0.50	> 0.40	Near-disconnect	X

? C3 Packet Loss at the Empirical Boundary

The largest published DSRC field study (CAMP VSC3, 2018) reports packet loss exceeding 25 % only at distances > 400 m in dense urban environments with heavy truck traffic. At typical intersection spacings of 200–400 m, a 30 % loss rate represents a worst-case condition that would already trigger alarm thresholds in a real Traffic Management Center.

i Missing — Correlated Loss Model

The thesis models packet loss as **i.i.d. Bernoulli trials** (p_{loss} per packet per receiver). In reality, vehicular channel losses are *spatially and temporally correlated*: if one packet is lost due to a fade event, subsequent packets within the coherence time (typically 5–50 ms at urban speeds) are also likely lost. This correlation significantly affects the proposed *historical state estimation* mechanism, which assumes that at least some recent messages arrive to establish a trend.

3.3 V2X Technology Characterization

The thesis (§ 6.4.1) states:

> *"DSRC (IEEE 802.11p): typical latency 50–100 ms, packet loss up to 10 %"*

? DSRC Characterization Is Incomplete and Potentially Misleading

- The 50–100 ms range applies to *V2V* at 200–300 m in urban settings. For *V2I* with fixed RSUs, typical latency is **10–50 ms** at the same range.
- "Packet loss up to 10 %" is cited without distance or conditions. At 100 m line-of-sight, DSRC loss is typically < 1 %. The 10 % figure applies at 250–350 m with partial obstructions.
- No mention is made of DSRC's *contention-based* channel access (CSMA/CA), which causes congestion-induced losses at high vehicle densities (> 200 veh/km/lane)—a critical factor for intersection communication.

For C-V2X, the thesis states:

> *"C-V2X (LTE-V2X, 5G NR-V2X): latency 20–50 ms, packet loss up to 5 %"*

? Conflation of LTE-V2X Mode 4 and 5G NR-V2X

These are **fundamentally different technologies**. LTE-V2X Mode 4 (vehicle-scheduled, no network) achieves 20–100 ms latency in field tests (5GAA 2021 report). 5G NR-V2X (3GPP Rel. 16+) can achieve < 10 ms but requires 5G NR base station coverage not yet widely deployed for V2X. Grouping them under a single range obscures this critical deployment distinction.

4 Traffic Simulation Fidelity

4.1 Simulator Identification Discrepancy

! Critical Inconsistency Across Document Versions

- `version3.pdf` (§ 4.1.1): states "*SUMO Traffic Simulator*"
- `report_2_version.pdf` and `report_2_version-1.pdf` (§ 4.2): state "*TRAFI simulator with the SAFE car-following model*"
- `report_1_version_tempo.pdf` (§ 3.3.1): discusses "SAFE model" benchmarking and mentions SUMO separately

SUMO **does not natively implement the SAFE** (Stochastic Automaton Follow-the-leader Engine) **car-following model**. SUMO's built-in models are Krauss, IDM, PWagner2009, and Wiedemann. SAFE is a separate model from the TRAFI simulation environment developed at ETH Zurich.

This discrepancy means either: (a) the authors intend a modified SUMO with a custom SAFE implementation (not described), or (b) the simulator choice was changed between versions without updating all sections. Neither interpretation reflects a mature, validated simulation setup.

4.2 Network Topology Realism

The thesis uses a synthetic 3×3 grid network with unspecified or varying parameters across versions (road segment length $\sim$300 m, 2–3 lanes per direction, 50 km/h speed limit).

? The 3×3 Grid Is Not Representative of Real Urban Networks

- Real urban networks contain T-intersections, asymmetric junctions (3-way, 5-way, 7-way), roundabouts, and merge/diverge points that fundamentally alter coordination dynamics.
- No pedestrian signals, transit priority, or turn restrictions are modeled—all of which significantly affect signal timing.
- The uniform 300 m spacing assumes a planned grid city (e.g., parts of Phoenix or Beijing), not an organically grown city such as **Dhaka**, where IUT is located. Dhaka's intersection spacing ranges from 50 m to 800 m with highly irregular geometry.
- Every intersection is identical, eliminating the coordination challenge of balancing arterial progression with local access.

4.3 Traffic Demand Characterization

The thesis (Table 4.1 in `report_2_version.pdf`) defines three demand levels: Low (400 veh/h/lane), Medium (800 veh/h/lane), High (1200 veh/h/lane).

i Demand Values Are Within Range but Lack Operational Realism

These values fall within realistic bounds for urban arterials (capacity at signalized intersections is typically 600–900 veh/h/lane per movement). However, several critical elements are absent:

- **No temporal variation**—no morning/evening peak patterns.
- **No origin–destination matrix**—flows appear uniform, eliminating route-choice effects.
- **No heavy-vehicle mix**—even 5 % trucks significantly affects queue discharge rates and gap acceptance.
- The "high demand" of 1200 veh/h/lane likely exceeds the capacity of a 3-lane approach with standard timing, creating gridlock—yet the thesis does not discuss implications for MARL learning stability.

5 Hardware and Software Architecture Accuracy

5.1 Claims in the Thesis

Section 6.4.2 lists five component categories: sensors, communication module, decision engine, actuation interface, and training infrastructure.

! Architecture Description Is Purely Taxonomic — No Actionable Specifications

- **No specific hardware platform** is identified (NVIDIA Jetson? Intel NUC? ARM-based TCU?). Inference latency varies by 10× across these platforms.
- **No neural network size** is specified. A QMIX agent with 128×128 hidden layers for 9 agents requires ~ 0.5 MB of parameters and achieves < 1 ms inference on a Jetson Nano—but this is never calculated.
- "GPU-accelerated edge devices" is **misleading**: most deployed traffic controllers use ARM CPUs without GPUs. GPU-equipped edge devices cost $500–2 000 per intersection, i.e., 5–20× the cost of a standard ATC. No cost analysis is provided.
- "NTCIP standard" is mentioned without specifying data objects (e.g., NTCIP 1202 v03) or the communication protocol (SNMP vs. RESTful API). NTCIP compliance for real-time phase control requires sub-100 ms response, which must be verified.
- The **cloud-to-edge deployment pipeline** (model packaging, OTA updates, version management, rollback) is entirely absent.

5.2 Real-World Reference: Surtrac

For calibration, the Surtrac system (Carnegie Mellon, deployed in Pittsburgh since 2012) employs:

- Sensing: radar + camera fusion per intersection
- Computation: standard PC (Intel i7, **no GPU**)
- Communication: **dedicated fiber mesh** (not V2X wireless)
- Actuation: direct interface to Econolite ASC/3 controllers
- Decision cycle: **1–5 seconds**

? 10-Second Decision Interval Is 2–10× Slower Than Deployed Systems

The thesis's 10-second decision interval is twice as slow as Surtrac's worst case and 10× slower than its best case. In congested conditions, a 10-second interval means the system **cannot respond to rapid queue buildup** between phases. The thesis neither justifies this choice nor analyzes its impact on performance.

6 Safety and Reliability Claims

6.1 Graceful Degradation Guarantee

The thesis (§ 3.7.1) asserts:

$$V_{\text{local}} \leq V(\mathcal{C}) \leq V_{\text{ideal}}$$

! Degradation Bound Is Unproven and Not Universally True

- The lower bound V_{local} assumes that the local-only policy *always* outperforms a mis-coordinated cooperative policy. This is **false**: partially-coordinated agents can produce *worse* outcomes than fully independent agents due to phase interference (e.g., two neighbors simultaneously granting green to conflicting flows based on different stale information).
- **No formal proof** is provided in any version of the thesis. The inequality appears as an assertion, not a theorem.
- Real-world safety standards (ISO 26262, IEC 61508) require *verified* safety properties. The thesis provides **no verification** whatsoever.

6.2 Local Fallback Claims

The thesis states that the system falls back to a "pre-timed or actuated plan" upon total communication failure.

? Missing — Fallback Transition Logic

In real traffic controllers, switching between adaptive and pre-timed modes requires:

- A **transition plan** to avoid signal phase jumps that could trap vehicles in intersections ("phase trap").
- **Minimum green time guarantees** maintained even during transition.
- **Coordination with adjacent intersections** to prevent "green wave" disruption.
- A **recovery plan** for returning to adaptive mode after communication restoration.

None of these are discussed. Improper mode transitions have caused safety incidents in deployed adaptive systems (e.g., the 2008 SCATS incident in Melbourne that produced a 45-minute gridlock).

7 Mathematical Formalization Accuracy

7.1 DEC-POMDP Formulation

The DEC-POMDP tuple $\langle \mathcal{N}, \mathcal{S}, \{A_i\}, \{O_i\}, T, R, \{\Omega_i\}, \gamma \rangle$ is standard and correctly stated.

✓ AoI-Augmented Observation Space — A Genuine Contribution

The communication-augmented observation $o_i^t = \langle s_{\text{local},i}^t,\ \{m_{j\to i}^{t-\tau_{ji}},\ \tau_{ji}\}_{j\in\mathcal{N}_i^{\text{rec}}}\rangle$ is a valid and well-motivated extension that makes Age of Information (AoI) explicit. This formulation aligns with the AoI literature (Kaul et al., 2012; Yates et al., 2021) and represents one of the thesis's genuine contributions.

7.2 QMIX Integration

✓ QMIX Formulation Is Correctly Stated

The QMIX formulation with monotonic mixing network $(\partial Q_{\text{tot}}/\partial Q_i \geq 0)$, hypernetworks, and TD loss function matches the original Rashid et al. (2018) paper. The freshness-weighted modification $Q_{\text{tot}} = f_{\text{mix}}(\mathbf{Q},\ s\odot\mathbf{w})$ with $w_i = e^{-\alpha\tau_i}$ is a reasonable extension, though unvalidated.

? Freshness Weighting Breaks the Monotonicity Guarantee

The original QMIX monotonicity constraint is proven for the standard mixing network. When the global state s is replaced by $s\odot\mathbf{w}$ (element-wise product with freshness weights), monotonicity of the hypernetwork outputs with respect to Q_i is **no longer guaranteed**, because the freshness weights are a function of the communication channel state, not the action values. The thesis does not address this theoretical implication.

7.3 Historical State Estimation

The linear extrapolation formula is given as:

$$\hat{s}_{\text{local},j}(t) \;=\; s_j(t_{\text{last}}) \;+\; \frac{t-t_{\text{last}}}{t_{\text{last}}-t_{\text{prev}}}\big(s_j(t_{\text{last}})-s_j(t_{\text{prev}})\big)$$

? Linear Extrapolation of Traffic State Is Unreliable

Traffic state variables (queue length, waiting time) are **non-monotonic and discontinuous**: a queue grows during red, shrinks during green, and exhibits step changes when a platoon arrives. Linear extrapolation from two past observations can produce absurd estimates (negative queue lengths, unbounded waiting times). The thesis mentions EMA in `version3.pdf` but provides no formula and does not analyze its interaction with the linear extrapolation.

8 Empirical Validation Status

! Single Most Critical Accuracy Concern — No Experiments Were Conducted

Chapter 5 is titled "Results and Analysis" and contains tables of performance metrics (average travel time, queue length, etc.) presented as experimental results. However:

1. Chapter 4 (§ 4.1) explicitly states the simulation is "intended for **future** empirical validation."
2. **No code repository, dataset, or replication instructions** are provided.
3. Performance values (e.g., "98.5 % retention at C2") appear with **no confidence inter-**

vals, no number of runs, and no statistical tests.

4. The "car-following model benchmarking" in § 5.2 compares SAFE, GIPPS, and IDM—but if the simulator is SUMO (which does not implement SAFE), **how were these results obtained?**
5. Standard deviations mentioned in `overall.pdf`'s description of `version3.pdf` **do not appear** in the actual thesis text of any examined version.

The honest characterization would be "**Projected Performance Analysis Based on Theoretical Reasoning**" rather than "Results and Analysis."

Table 5. **What Chapter 5 Claims vs. What It Actually Represents**

Element	Thesis Presentation	Actual Status
Performance tables	Measured results	Theoretical projections
Degradation curves	Observed data	Logistic model fitted to assumed parameters
Comparative analysis	Method comparison	Qualitative reasoning about expected behavior
Key findings	Experimental conclusions	Hypotheses about system behavior
Standard deviations	Statistical measures	Not present in any examined version

9 Reward Design Practicality

The thesis proposes a cooperative reward combining local and global components but does not specify exact weighting in the final version.

? Unresolved Reward Weighting Problem

In cooperative MARL for traffic control, the relative weight of local vs. global reward fundamentally determines learned behavior:

- **High global weight** → agents sacrifice local performance for network benefit (e.g., one intersection absorbs longer queues to help neighbors), which can cause citizen complaints at the "sacrificed" intersection.
- **High local weight** → agents behave nearly independently, negating cooperation benefits.
- The optimal weight is **scenario-dependent** and typically requires extensive per-network tuning.

No version provides actual weight values or analyzes sensitivity to this critical parameter. In real deployments (e.g., Surtrac), reward weights are calibrated per-intersection based on traffic engineering judgment, not learned end-to-end.

10 Scalability Claims

? 9 Intersections Do Not Support Scalability Claims

A 3×3 grid is the *minimum non-trivial* multi-agent scenario. Real urban deployments involve 50–500+ intersections. Key unaddressed scalability issues include:

- **Communication graph diameter:** In a 100-node network, information from distant intersections arrives with *compounded latency*, a phenomenon absent in 9-node grids.
- **Mixing network scaling:** The QMIX mixing network input size scales linearly with n. For $n = 100$, the hypernetwork output dimensions may exceed GPU memory.
- **Curriculum training cost:** The 4-phase curriculum requires sequential training under increasing communication stress. For large networks, each phase may require weeks on multi-GPU setups. **No computational cost analysis** is provided.

11 Comparison with Deployed Systems

Table 6. Thesis Claims vs. Deployed System Reality

Aspect	Thesis Claim	Deployed Reality	Gap Assessment
Decision interval	10 s	1–5 s (Surtrac)	2–10× slower
Sensing	Not specified	Radar + camera fusion	Major gap
Communication	V2X wireless	Fiber mesh (Surtrac)	Different paradigm
Scale tested	9 intersections	50+ (Surtrac)	5× fewer
Validation	None (theoretical)	3+ years field data	No comparison possible
Safety certification	Not discussed	Required by local DOT	Critical gap
Cost / intersection	Not estimated	\$5 K–\$50 K	No feasibility analysis
Weather robustness	Not discussed	Major failure mode	Critical gap

12 Accuracy of Self-Assessment Claims

The thesis makes six contribution claims. Table 7 evaluates each for self-assessment accuracy.

Table 7. **Contribution Claim Accuracy Assessment**

Claim	Accurate?	Assessment
1. Theoretical framework for communication imperfection modeling	✓	Genuine contribution; AoI-augmented DEC-POMDP is novel for traffic MARL.
2. Latency-resilient design principles	!	Principles stated but not validated—these are hypotheses, not contributions.
3. Structured comparative framework	!	Framework exists on paper but was never executed; a plan, not a contribution.
4. Cooperative reward design	!	Described conceptually; no specific formula, weights, or ablation study provided.
5. Theoretical analysis of comm.–performance relationships	!	Qualitative reasoning with a logistic model fitted to *assumed* (not measured) parameters.
6. Bridging theory and practice	X	No practical validation exists. The thesis bridges theory to *simulation design*, not to practice.

13 What the Thesis Gets Right

Despite the concerns documented above, several aspects of the thesis are genuinely accurate and valuable:

✓ Strengths of the Released Thesis

Problem identification is excellent. The gap between ideal communication assumptions in MARL traffic control literature and real-world deployment conditions is real, important, and under-explored. The thesis correctly identifies this as a critical barrier.

Literature review is thorough and well-organized. The thematic progression (single-agent → MARL → communication-aware → gap analysis) is logical and covers key works accurately.

The AoI-augmented observation formulation is sound. Making information freshness explicit in the DEC-POMDP observation space is both theoretically clean and practically relevant.

The four-configuration experimental design (if executed) would be meaningful. The progression from ideal to extreme provides a structured way to characterize degradation—a dimension missing from the literature.

The curriculum training concept is well-motivated. Gradually increasing communication stress during training aligns with established curriculum learning principles in RL.

14 Recommendations

Recommendations are prioritized from P0 (essential for basic accuracy) to P3 (desirable for completeness).

Table 8. Prioritized Recommendations for Improving Real-World Accuracy

Priority	Recommendation	Rationale
P0	Relabel Chapter 5 as "Projected Performance Analysis" rather than "Results and Analysis"	Prevents systematic reader misinterpretation; addresses the most critical accuracy concern
P0	Resolve the SUMO vs. TRAFI/SAFE discrepancy explicitly	Fundamental to any claim of reproducibility
P1	Replace C4 with a realistic extreme scenario ($\tau \in [200, 500]$ ms, $p_{\text{loss}} = 20\,\%$)	Removes artificial inflation of resilience benefit
P1	Add a non-zero baseline to C1 ($\tau \in [5, 20]$ ms)	Reflects minimum physically realistic latency
P1	Specify the exact reward function with numerical weights	Essential for reproducibility and practical relevance
P2	Add a correlated packet loss model option	Significantly affects the resilience mechanism's effectiveness
P2	Justify the 10-second decision interval or reduce to 5 seconds	Aligns with deployed-system standards (Surtrac: 1–5 s)
P2	Provide computational cost analysis (training time, inference latency per agent)	Essential for any deployment feasibility discussion
P3	Discuss weather, sensor failure, and maintenance scenarios	These are the dominant real-world failure modes for ITS
P3	Add a cost-per-intersection estimate	Required for any practical deployment conversation

15 Conclusion

The released thesis presents a **theoretically sound but empirically unvalidated** framework. Its real-world accuracy is undermined not by fundamental theoretical errors but by five systemic issues:

1. **Presentation of projections as results**—the most serious accuracy concern, affecting the credibility of the entire findings chapter.
2. **Simulator inconsistencies** that fundamentally undermine reproducibility.
3. **Unrealistic extreme scenarios** (C4) that artificially inflate the apparent value of the contribution.
4. **Superficial deployment discussion** lacking the specificity needed for any real implementation effort.
5. **Unproven safety claims** that could be dangerous if taken at face value by practitioners.

The thesis would be **significantly more accurate** if it explicitly positioned itself as a *theoretical framework with a simulation design for future validation*, relabeled Chapter 5 honestly, and replaced the deployment discussion with a candid assessment of what remains to be done before real-world testing is possible.

As it stands, the thesis makes a **genuine theoretical contribution**—the AoI-augmented DEC-POMDP for traffic MARL—but overstates its practical readiness by a considerable margin. The central accuracy deficit is the gap between the thesis's implicit claim of "*we have developed and evaluated a deployable system*" and the reality of "*we have designed a theoretical framework and a plan for testing it.*"

This assessment is based solely on the content of the provided documents and published empirical data. No independent experimental verification was conducted.